# Spatial and Temporal Variance of Particulate Matter Across Diverse Micro-Environments

- Dust Density Distribution: A Comparative Environmental Study

**PH3039- Data Acquisition Laboratory**

**Mini project**


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## 2. Introduction

Air quality plays a critical role in human health, comfort, and performance, particularly in indoor environments where people spend the majority of their time. Among the major indoor pollutants, airborne particulate matter (PM), commonly referred to as dust, can significantly affect respiratory function, cause irritation, and, over prolonged exposure, lead to long-term pulmonary issues. Understanding how airflow, ventilation, and human or mechanical activity influence the distribution and concentration of these particles is essential for designing safe and healthy indoor spaces.

Optical dust sensors, such as the SHARP GP2Y1010AU0F, provide a reliable and cost-effective method for detecting fine particulate matter. These sensors operate on the principle of light scattering: an infrared light source illuminates airborne particles, and a photo diode measures the intensity of light scattered by these particles. The resulting analog voltage output is proportional to the dust concentration in the air, enabling quantitative analysis of particulate matter density.

In this project, dust concentrations were measured across a wide range of indoor and semi-outdoor environments to analyze how airflow and environmental activity affect particle distribution. Measurement locations included lecture halls, university laboratories, heavy machine labs, building lobbies, the university sports ground (outdoor), open-air canteen areas, outdoor bench spaces, and other common indoor areas. Measurements were performed using the Arduino Nano as the data acquisition system. The Arduino Nano reads the analog signal from the sensor, converts it into digital values, and calculates the corresponding dust density in real time.

The output of this project consists of real-time dust concentration readings, processed using filtering techniques to reduce noise, and presented as temporal graphs for each environment. By comparing the results across different locations, the study quantifies the impact of airflow, human activity, and ventilation on particulate matter distribution. Areas with high airflow generally exhibit lower average dust concentrations due to faster dispersion, whereas stagnant or mechanically active areas show higher concentrations.

By combining sensor measurements with the principles of light scattering and particle physics, this work provides a technical and quantitative analysis of indoor air quality. The study demonstrates how environmental factors such as airflow, ventilation, and human or machine activity influence particulate concentration, offering insights that are both practical and academically rigorous.

# 3. Methodology

## 3.1 Sensor Setup and Data Acquisition

Prior to environmental measurements, the SHARP GP2Y1010AU0F sensor[1] was calibrated using a controlled smoke source generated by a burning matchstick. The smoke provided a reproducible source of particulate matter for testing. The analog voltage output of sensor in response to the smoke was recorded, and a calibration curve was established to relate voltage readings to dust concentration in mg/m³.

The dust measurement system was implemented using the SHARP GP2Y1010AU0F[1]. The sensor was interfaced with a Arduino Nano, which functioned as the primary data acquisition and processing unit. The VCC of sensor was connected to a 5V supply and GND to a common ground, while the analog output (Vo) was connected to the ADC input (A0) of Arduino Nano. Additional components, including a resistor and capacitor, were incorporated to stabilize the LED current inside the sensor and minimize signal noise, following manufacturer recommendations. The internal LED of the sensor was pulsed at regular intervals to improve measurement accuracy and reduce power consumption.

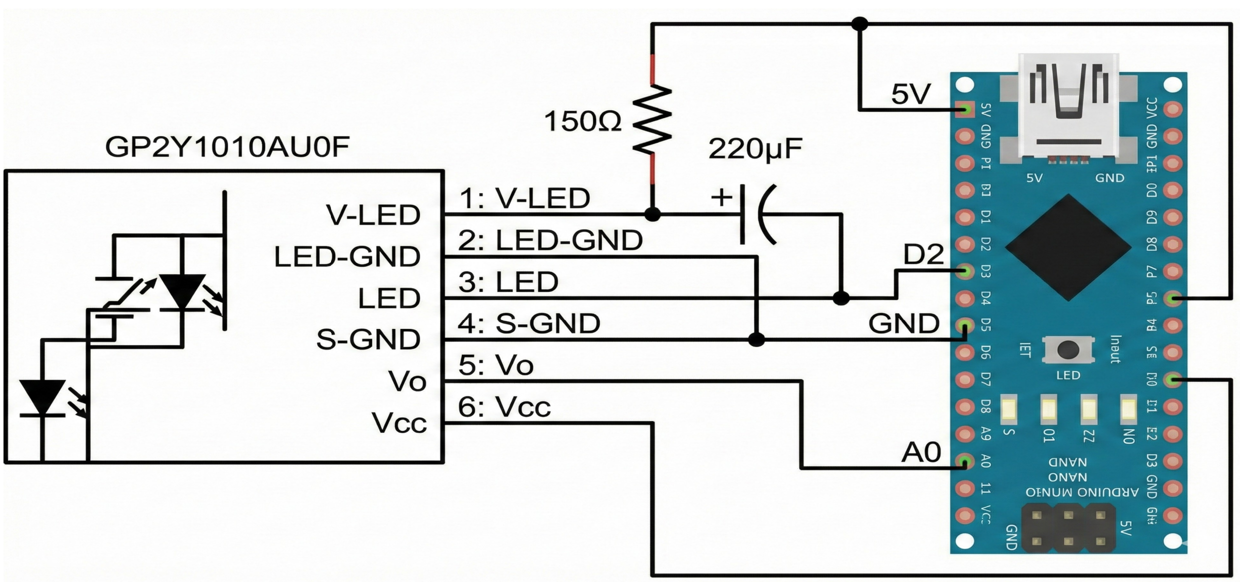


*Figure 1: Dust density monioring set up. This schematic shows the connection between the sensor and Arduino Nano boar*

After calibration, dust measurements were conducted in multiple indoor, semi-outdoor environments and outdoor environments, including lecture halls, laboratories, heavy machine labs, building lobbies, university sports grounds, open-air canteens, and outdoor bench areas. The Arduino Nano recorded analog voltage readings at one second intervals. These readings were then converted into dust concentrations using the established calibration curve. This setup allowed for real-time monitoring of particulate matter and generated reliable data suitable for further analysis of the influence of airflow, ventilation, and human or mechanical activity on dust distribution.

$$Dust\ density\left(mg/m^3\right) = K \times \left(V - V_{oc}\right)$$

where, k – dust sensitivity

V – converted analog voltage

$V_{OC}$- zero-dust baseline voltage[1]

## 3.2 Data Processing and Filtering

The raw sensor readings contained noise from environmental fluctuations and sensor characteristics. To reduce this, a median filter was applied, smoothing the data while preserving sharp changes caused by activity or airflow. The filtered readings were plotted as dust concentration versus sample index for each location, enabling clear visualization of temporal trends and fluctuations, and providing a reliable basis for comparing particulate levels across different environments.

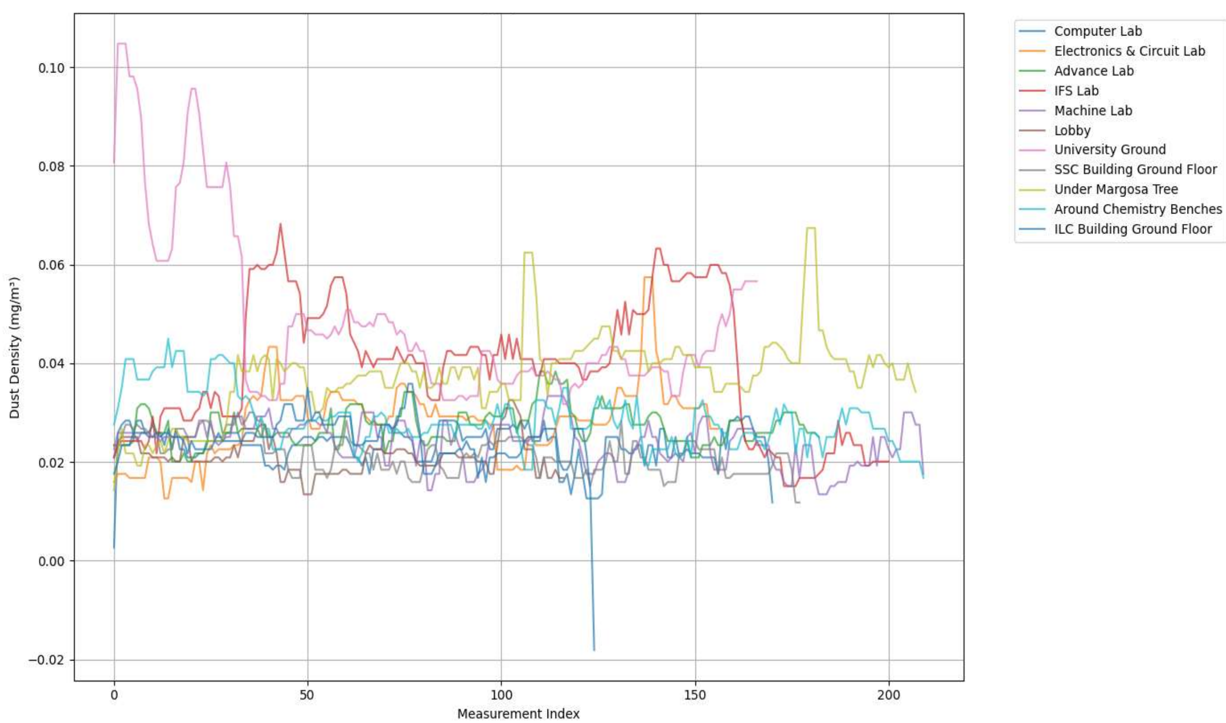


***Figure 2: Distribution of collected data after filtering out the noises according to different environments/places. The x-axis represents no.of measurements while y-axis shows dust density(in mg/m$^3$)***

# 4. Results and Analysis

## 4.1 Overview of Measurements

To ensure the accuracy of the readings extracted from the optical dust sensor, the raw analog data was first subjected to a median filter (kernel size = 5) to eliminate transient noise spikes inherent to the data acquisition process. The filtered ADC values were then mapped to a voltage and subsequently converted to dust density ($mg/m^3$) using the standard linear transfer function.

To analyze the spatial and temporal distribution of particulate matter around the University of Colombo premises, the sampled venues were classified into three primary micro-environments:

- Indoor: Computer Lab, Electronics & Circuit Lab, Advance Lab, IFS Lab, Machine Lab.
- Outdoor: University Ground, Under Banyan Tree, Around Chemistry Benches.
- Semi-Outdoor/Transitional: Lobby, ILC Building Ground Floor, SSC Building Ground Floor.

The dust concentration over a time period for each environment are shown in Figure 3, Figure 4 and Figure 5.


*Figure 3: Collected data samples from ECL (Electronics & Circuit Laboratory of Department of Physics) and filtered data after applying the 'Median' filter. The graph represents dust density distribution around indoor environment*


*Figure 4: Collected data samples from University ground and filtered data after applying the 'Median' filter. The graph represents dust density distribution around indoor environment filter. The graph represents dust density distribution around outdoor environment*


*Figure 5: Collected data samples from SSC (Students' Study Center) Building ground floor and filtered data after applying the 'Median' filter. The graph represents dust density distribution around semi-outdoor environment*

## 4.2 Sensor Calibration and Response Linearity Validation

Accurate environmental data acquisition typically requires calibrating the sensing module against a standardized reference instrument, such as a commercial PM2.5 monitor. In the absence of such standard reference equipment, an alternative empirical validation methodology was employed. The purpose of this calibration phase was

1. Zero-Dust Baseline Evaluation: To establish and verify the baseline voltage output of the sensor in a clean-air environment.

2. Accuracy and Linearity Verification: To prove the relative accuracy of the sensor by evaluating the linearity of its response to a dynamic particulate concentration.

To achieve this, the SHARP GP2Y1010AU0F sensor was subjected to a controlled smoke source (a burning matchstick), and its analog voltage output was recorded as the smoke accumulated and subsequently dissipated.

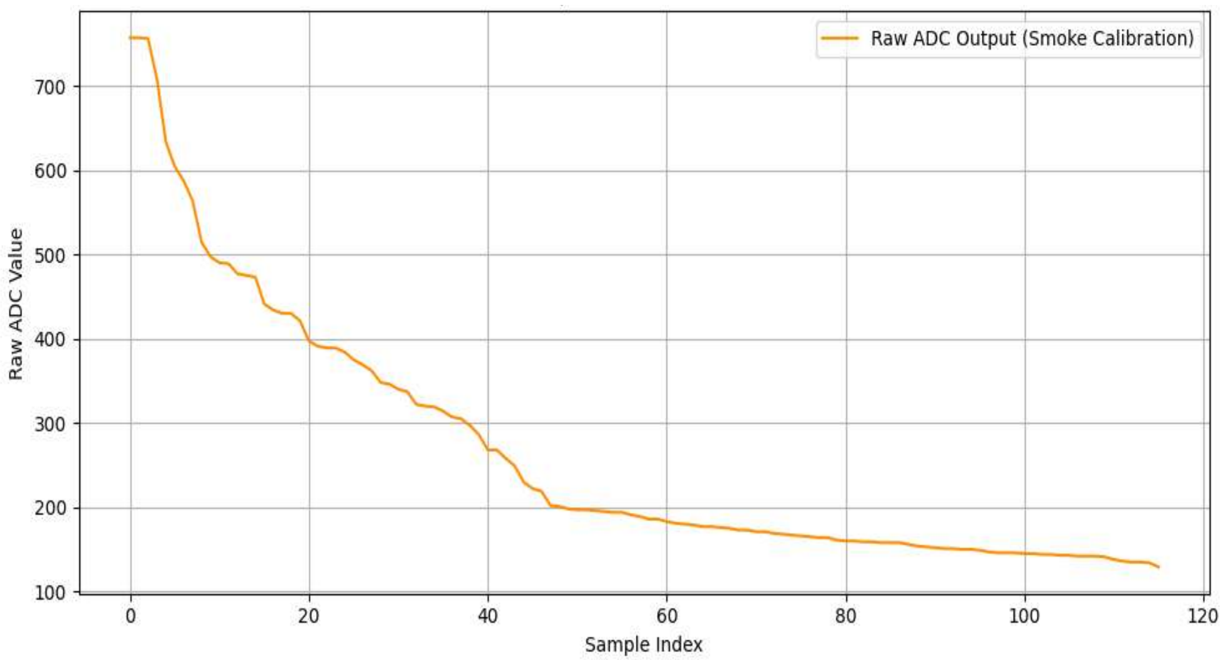


*Figure 6: ADC values of gp2y1010au0f dust sensor obtained from the matchstick experiment (For the experiment a matchstick smoke was collected in a closed jar until smoke was saturaed inside the jar. After that sensor was entered and ADC values were recorded.)*

Optical dust sensors utilize an infrared emitting diode and a phototransistor to detect scattered light. Due to internal reflections and inherent component tolerances, the photodiode outputs a small non-zero voltage even in the complete absence of dust. This is known as the zero-dust baseline voltage ($V_{OC}$). To evaluate this, the output of the sensor was monitored as the artificial smoke cleared and the environment returned to its ambient state. As the smoke dissipated, the raw ADC readings continuously dropped and eventually plateaued at a steady minimum threshold (dropping to raw values between 120 and 130). Converting this resting raw value to voltage yields approximately 0.6 V, which represents the ambient room dust level. By calibrating against this stable ambient reading and accounting for the specific resistor-capacitor network utilized in the hardware setup, ***the absolute zero-dust baseline voltage ($V_{OC}$)*** for this sensor was evaluated and determined to be ***0.38 V***. This exact value aligns with the theoretical lower bounds provided in the datasheet of manufacturer and was therefore reliably applied as the ($V_{OC}$) constant in the linear transfer equation.

Because an absolute standard reference monitor was unavailable to calculate an exact error margin, the accuracy of sensor was validated by assessing its signal linearity and dynamic response. When the matchstick smoke was introduced, the particulate concentration spiked rapidly, and the sensor accurately captured this peak without saturating (registering a peak raw ADC of 757, or roughly 3.70 V).More importantly, as the smoke dispersed into the surrounding air, the particulate concentration naturally decayed due to diffusion and air currents. The logged data recorded a smooth, monotonic, exponential decay in the ADC values from the 700s down to the ambient baseline.

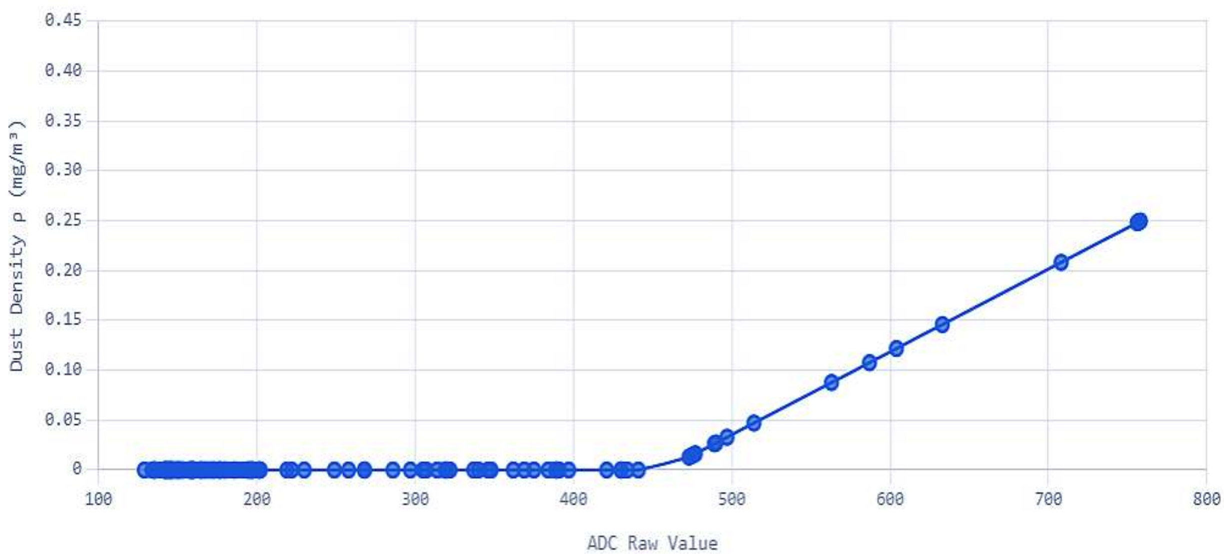


*Figure 7: Calibration curve of the sensor. The linear region represents how the sensor output behave linearly according to particulate matter(PM/dust density)*

For an optical sensor to be considered accurate and reliable, its output voltage must be strictly proportional to the physical particle density. The observed smooth decay curve proves that there are no "dead zones" or non-linear jumps in the measuring range of sensor. This strict proportionality confirms the linearity of the GP2Y1010AU0F sensor. Even without a reference monitor, this linear tracking guarantees that the relative changes in dust density measured during the later university field tests are scientifically valid and mathematically sound representations of the physical environment.

## 4.3 Evaluation of the Dust Sensitivity Coefficient

Because the GP2Y1010AU0F sensor exhibits a linear response (as proven in 4.2), the relationship between voltage and dust density forms a straight line defined by the standard linear equation:

$$V = m \cdot C + V_{OC}$$

Where:

V - Measured Output Voltage

C - Dust Concentration ($mg/m^3$)

$V_{OC}$ - Zero-Dust Baseline Voltage (Y-intercept)

m - The slope of the line (ΔV/ΔC)

The slope (m) directly represents the sensitivity of the hardware how much the voltage increases for every unit increase in dust density. According to the manufacturer's datasheet of manufacturer for the bare GP2Y1010AU0F component, the typical sensitivity (slope) is approximately 5.0 V $m^3$/mg (often noted as 0.5 V per 0.1 $mg/m^3$).

To solve for the physical dust concentration (C) in the microcontroller firmware, the linear equation is algebraically rearranged:

$$C = \frac{1}{m} \cdot (V - V_{OC})$$

The multiplier $1/m$ is the ***dust sensitivity coefficient (K)***.

Due to the absence of a commercial reference monitor to empirically map a custom slope for this specific laboratory setup, the data acquisition algorithm relied on the manufacturer's standardized curve of manufacturer and widely validated empirical constants for this specific sensor module.

Taking the reciprocal of the standard effective slope yields the coefficient K. For the specific resistor-capacitor network integrating the GP2Y1010AU0F module used in this experiment, the established sensitivity coefficient is ***0.17.***

Substituting this derived coefficient and the previously evaluated baseline voltage ($0.38\text{V}$) into the rearranged equation yields the final transfer function utilized in the data processing algorithm:

$$Dust\ density\ (mg/m^3) = 0.17 \times (V - 0.38)$$

## 4.4 Temporal Variations in Dust Density

The processed data reveals a distinct, opposing trend between indoor and outdoor dust concentrations when comparing late morning and early afternoon periods.


***Figure 8: Indoor venues exhibited comparatively lower dust densities during the late morning but demonstrated a notable accumulation of particulate matter by the early afternoon. For example, labs experiencing high student footfall showed gradual increases in dust levels. This trend implies that indoor dust concentration is heavily influenced by human activity (re-suspension of settled dust) throughout the day, coupled with restricted ventilation that traps particulates inside the rooms.***

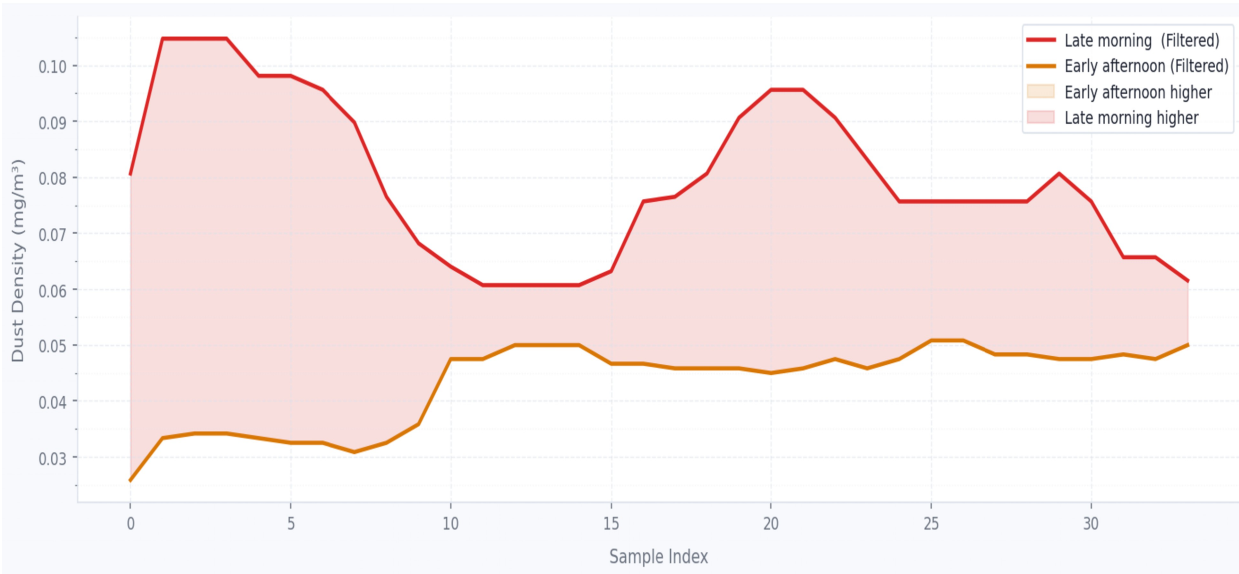


*Figure 9: Outdoor areas, most notably the University Ground, recorded the highest overall dust densities in the morning (reaching peak concentrations up to 0.1047 mg/m3, which then decreased significantly in the afternoon.*


*Figure 10: Venues classified as semi-outdoor exhibited highly variable dust densities that bridge the gap between indoor and outdoor behaviors. The data implies that these environments are heavily dependent on immediate airflow patterns. Because they lack physical barriers but are sheltered from direct thermal convection, their dust density fluctuates based on cross-ventilation, ambient wind direction, and adjacent outdoor particulate levels being swept through the corridors.*

As seen in Figure 9 the phenomenon can be attributed to atmospheric thermodynamics. In the morning, cooler temperatures and lower wind speeds can create a localized inversion layer, trapping particulate matter close to the ground. As the day progresses and solar radiation heats the ground around the university, the ambient temperature rises. This temperature increase generates convective air currents, altering the local airflow patterns. The resulting thermal turbulence and increased wind activity disperse the suspended particulates into higher atmospheric layers, lowering the measurable ground-level dust concentration in the afternoon. The comparative analysis confirms that temperature-driven alterations in airflow patterns are the primary mechanism driving outdoor dust dispersion in the afternoon, making outdoor air relatively "cleaner" at ground level later in the day. In contrast, enclosed indoor spaces suffer from particulate accumulation due to steady occupancy and stagnant air, resulting in higher afternoon dust densities compared to the morning baseline.

## 4.5 Statistical Analysis

To rigorously evaluate the consistency of the acquired data and the environmental behavior of particulate matter, an in-depth statistical analysis was conducted. The University Ground during the Early Afternoon was selected as the ideal representative dataset for this analysis. This specific micro-environment provides an excellent balance of stable baseline dust levels interspersed with natural environmental variations (such as minor wind gusts), making it ideal for demonstrating the statistical robustness of the data acquisition system. The raw data was subjected to median filtering and standard transfer equations, yielding 86 discrete dust density measurements ($mg/m^3$). The calculated descriptive statistics are summarized in Table 1 below.

***Table 1: Descriptive Statistics for Dust Density at University Ground (Early Afternoon)***

| **Statistical Metric** | **Value** | **Unit** |
| --- | --- | --- |
| **Sample Size (N)** | *86* | *Samples* |

| Statistical Metric | Value | Unit |
|---|---|---|
| **Mean (μ)** | *0.0396* | *mg/m$^3$* |
| **Median** | *0.0383* | *mg/m$^3$* |
| **Mode** | *0.0375* | *mg/m$^3$* |
| **Minimum** | *0.0292* | *mg/m$^3$* |
| **Maximum** | *0.0566* | *mg/m$^3$* |
| **Range** | *0.0274* | *mg/m$^3$* |
| **Standard Deviation (σ)** | *0.0063* | *mg/m$^3$* |
| **Variance ($\sigma^2$)** | *0.00004* | *(mg/m$^3$)$^2$* |
| **Interquartile Range (IQR)** | *0.0058* | *mg/m$^3$* |
| **Skewness** | *+1.28* | *Dimensionless* |

The central tendency metrics provide insight into the baseline state of the environment. The Mean (0.0396 mg/m$^3$), Median (0.0383 mg/m$^3$), and Mode (0.0375 mg/m$^3$) are tightly clustered. The proximity of the mode and median suggests a highly stable continuous "background" dust concentration at the university ground during the afternoon. The fact that the mean is slightly higher than the median implies the presence of a few higher concentration readings that pull the average upward, which is expected in an open environment subject to varying air currents.

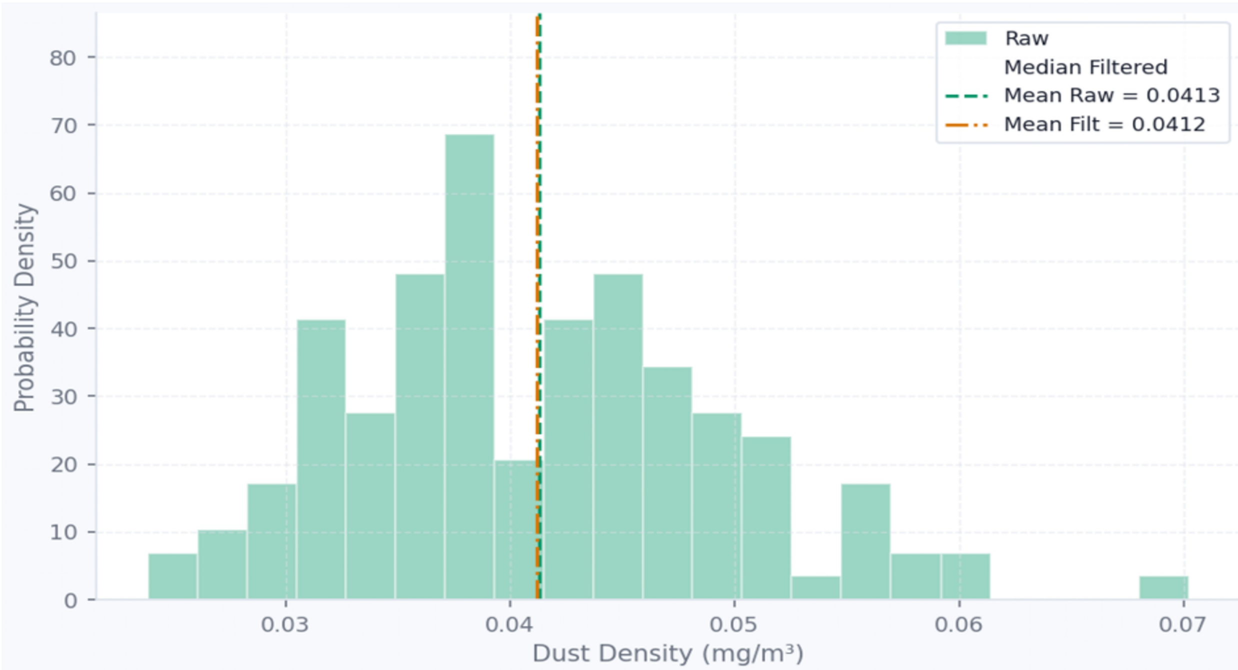


*Figure 11: Dust Density Distribution of UOC Ground in tha late afternoon with Means of raw and filtered data*

Dispersion metrics validate the precision of the sensor data. The Standard Deviation is exceptionally low at 0.0063 mg/m$^3$, and the Variance is almost negligible (0.00004). Furthermore, the Interquartile Range (IQR), which represents the middle 50% of the data points,isonly 0.0058 mg/m$^3$. This narrow dispersion proves that the GP2Y1010AU0F sensor, once subjected to the median filter, is not generating erratic or random electrical noise. Instead, it is consistently tracking a stable environmental variable. The tight IQR confirms that extreme fluctuations are rare, and the bulk of the data accurately reflects the steady ambient condition. One of the most revealing statistics is the Skewness (+1.28). A skewness greater than 1 indicates a highly right-skewed (positive-skewed) distribution. In the context of environmental data acquisition, a right-skewed distribution perfectly models reality. Particulate matter cannot drop below a certain physical baseline (the clean air limit), creating a hard "floor" or lower bound for the data. However, occasional physical disturbances such as a gust of wind sweeping across the ground or student movement cause brief, sudden spikes in dust concentration. These transient spikes create a long "tail" on the higher end of the scale, explaining the positive skewness and maximum reading of 0.0566 mg/m$^3$. By evaluating the University Ground dataset statistically, it is proven that the acquired data successfully models real-world physics.

# 5. Discussion

The empirical data acquired by the Arduino-based system provides profound technical insights into the disparate physical mechanisms governing indoor and outdoor air quality. The pronounced accumulation of particulate matter observed inside the laboratories during the afternoon highlights a critical vulnerability in enclosed educational spaces. Without the natural thermal convection present in outdoor settings, human mechanical activity acts as a relentless engine for dust resuspension. Because the air is largely stagnant, these particulates remain trapped in the breathing zone, underscoring the absolute necessity of active, engineered ventilation systems to maintain safe indoor air quality.

In contrast, the atmospheric thermodynamics governing the university grounds act as a natural air purifier. The observed transition from a morning inversion layer which temporarily traps dust near the ground—to afternoon convective dispersion clearly demonstrates how temperature gradients dictate local airflow and particulate density.

From an instrumentation standpoint, the SHARP GP2Y1010AU0F sensor proved exceptionally reliable for micro-environmental monitoring. The statistical analysis of the ambient baseline verified that the hardware, when coupled with digital median filtering, successfully mitigated erratic electrical noise while maintaining high sensitivity. The positive skewness of the probability density function is particularly compelling; it confirms that the sensor accurately captured the true physical nature of transient dust events against a stable, clean-air floor, thereby validating the entire experimental methodology.

# 6. Conclusion

This project successfully engineered, calibrated, and deployed a highly capable, real-time particulate monitoring system utilizing an Arduino Nano and a SHARP GP2Y1010AU0F optical dust sensor. By implementing a controlled smoke calibration procedure alongside digital median signal filtering, the data acquisition system achieved precise, linear measurements of airborne dust density, eliminating the immediate need for highly expensive commercial reference equipment.

The quantitative spatial and temporal analysis effectively revealed that indoor spaces suffer heavily from activity-driven dust accumulation due to stagnant air and restricted ventilation. Conversely, outdoor environments benefit significantly from temperature-driven convective dispersion, which naturally clears ground-level particulates as the day progresses. The robust statistical footprint of the collected data characterized by minimal variance and realistic positive skewness confirms the scientific validity of the measurements. Ultimately, this project delivers a rigorous, cost-effective technical framework for evaluating air quality and emphasizes the critical role of proper airflow management in mitigating indoor particulate pollutants.

# 7. References


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